

ELECTRIC VEHICLE TECHNOLOGY INTERNSHIP

TASK 2: BATTERY ANALYSIS

Submitted by	Program
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Objective

To study lithium-ion battery packs used in electric vehicles, calculate basic battery capacity and energy requirements, compare two battery configurations, and explain the importance of a Battery Management System (BMS).

Battery Analysis • Capacity • Energy • Configuration • BMS

1. Introduction to EV Batteries

The traction battery is the main energy-storage unit of a battery electric vehicle (BEV). It stores electrical energy and supplies it to the power electronics and electric motor. Modern EV battery packs are generally built from rechargeable lithium-ion cells because they offer a useful combination of energy density, power capability, efficiency, cycle life and relatively low self-discharge.

A complete battery pack is more than a collection of cells. It normally includes cells/modules, electrical interconnections, contactors and protection devices, temperature sensors, current/voltage measurement circuits, cooling or thermal-management hardware, and a Battery Management System (BMS).

1.1 Basic Battery Terminology

Term	Meaning
Cell	Smallest electrochemical unit that stores energy and produces a nominal voltage.
Module	Group of cells connected and mechanically packaged together.
Battery Pack	Complete high-voltage assembly containing cells/modules and supporting electrical, mechanical and thermal systems.
Voltage (V)	Electrical potential. Series-connected cells increase voltage.
Capacity (Ah)	Amount of electrical charge the battery can deliver under specified conditions.
Energy (Wh/kWh)	Approximate stored electrical energy, commonly estimated as voltage × capacity.

1.2 Series and Parallel Cell Connections

Cells are connected in **series** to obtain the required pack voltage and in **parallel** to increase capacity and current capability. A notation such as 96S2P means 96 cells/groups in series and 2 parallel strings.

2. Battery Capacity and Energy Calculations

For a simplified battery-pack calculation, nominal energy can be estimated from pack voltage and capacity. Practical usable energy is lower or different depending on operating limits, temperature, efficiency, aging and battery-control strategy.

2.1 Core Formulae

Battery energy: $E \text{ (Wh)} = V \text{ (V)} \times C \text{ (Ah)}$

Energy in kWh: $E \text{ (kWh)} = V \times C / 1000$

For identical cells: $V_{\text{pack}} = N_s \times V_{\text{cell}}$ and $C_{\text{pack}} = N_p \times C_{\text{cell}}$

2.2 Example Calculation

Assume an EV battery pack has a nominal voltage of **400 V** and a capacity of **100 Ah**.

$E = 400 \times 100 = 40,000 \text{ Wh} = 40 \text{ kWh}$

Therefore, the nominal stored energy is approximately **40 kWh**. If an illustrative 90% usable-energy assumption is used, estimated usable energy is $40 \times 0.90 = \mathbf{36 \text{ kWh}}$. This is an educational estimate, not a manufacturer-rated value.

2.3 Power and Current Example

If a 400 V pack supplies 40 kW of electrical power:

$I = P / V = 40,000 / 400 = 100 \text{ A}$

This demonstrates why EV battery design considers both energy capacity (kWh) and power/current capability. Actual current changes with acceleration, speed, road grade, efficiency and operating conditions.

3. Comparison of Two Battery Configurations

Consider a representative lithium-ion cell with a nominal voltage of **3.7 V** and capacity of **50 Ah**. These configurations are simplified examples for engineering learning.

Parameter	Configuration A 96S1P	Configuration B 96S2P
Cells in series	96	96
Parallel strings	1	2
Total cells	96	192
Nominal voltage	$96 \times 3.7 = 355.2 \text{ V}$	$96 \times 3.7 = 355.2 \text{ V}$
Capacity	$1 \times 50 = 50 \text{ Ah}$	$2 \times 50 = 100 \text{ Ah}$
Nominal energy	$355.2 \times 50 = 17.76 \text{ kWh}$	$355.2 \times 100 = 35.52 \text{ kWh}$

3.1 Interpretation

Both configurations have the same nominal voltage because they use the same 96 series cells. The 2P configuration doubles the pack capacity and nominal energy because two parallel strings store more charge. Parallel strings can also share current when the cells, interconnections and thermal system are properly designed.

Series connection mainly increases voltage, while parallel connection mainly increases capacity and current capability. A real EV pack is selected using the motor/inverter voltage range, required energy, peak power, thermal limits, packaging, cost, safety and cell characteristics.

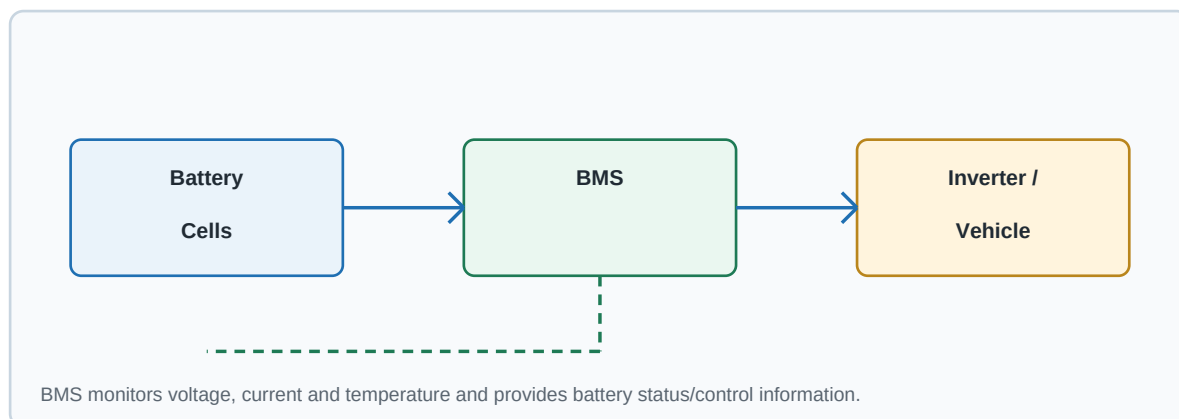
4. Battery Management System (BMS)

A BMS is the electronic monitoring and control system that helps keep the battery within safe operating limits while supporting performance and battery life. It monitors electrical and thermal conditions and can control contactors, charging limits, balancing and fault responses.

4.1 Major BMS Functions

Function	Purpose
Cell voltage monitoring	Checks individual cell/group voltages and identifies abnormal conditions.
Temperature monitoring	Tracks cell and pack temperature to help prevent operation outside limits.
State of Charge (SOC)	Estimates remaining charge using measured current, voltage and battery models.
State of Health (SOH)	Estimates battery condition and capacity degradation over service life.
Cell balancing	Reduces voltage differences between cells so the pack operates more uniformly.
Protection & fault handling	Can limit charging/discharging or open contactors when abnormal conditions are detected.

4.2 Simplified Battery-BMS Relationship



The BMS is therefore an essential part of a high-voltage EV battery system. It helps coordinate safe operation of the cells with the charger, inverter and vehicle control system.

5. Conclusion

This study shows how an EV battery pack is built from cells and how series/parallel connections determine its electrical characteristics. The relationship $E = V \times Ah$ provides a useful first estimate of stored energy, while practical battery design must also consider current capability, thermal management, efficiency, operating limits and aging.

The 96S1P and 96S2P comparison demonstrates that increasing the number of parallel strings increases capacity and nominal energy while keeping the same series voltage. The BMS plays a central role in monitoring, protection, balancing and estimation of battery condition, making it an important part of a safe and reliable EV energy-storage system.

Key Takeaways

- Series-connected cells increase pack voltage.
- Parallel-connected cells increase capacity and current capability.
- Battery energy is commonly estimated from voltage $\times$ capacity.
- BMS functions include monitoring, protection, balancing and battery-state estimation.
- Real EV packs require thermal, electrical, mechanical and safety design beyond simplified calculations.

References

1. U.S. Department of Energy, Alternative Fuels Data Center (AFDC) — general electric-vehicle and battery technology information.
2. M. H. Rashid, *Power Electronics: Circuits, Devices and Applications* — power-electronics fundamentals relevant to EV energy conversion.
3. Gregory L. Plett, *Battery Management Systems* — battery modeling and BMS concepts.
4. Numerical examples in this report are illustrative calculations prepared for internship learning; they are not manufacturer specifications.

End of Task 2: Battery Analysis